Parallel and distributed Computing

Lecture 27, 28, 29,30

Bazam-e-Jahan

Lecturer

The University of Faisalabad

Lecture 27
Scheduling

Scheduling

- Scheduling can be defined as "prescribing of when and where each operation necessary to manufacture the product is to be performed."

OR

- It is also defined as "establishing of times at which to begin and complete each event or operation comprising a procedure". The **principle aim** of scheduling is to plan the sequence of work so that production can be systematically arranged towards the end of completion of all products by due date.

Principles of Scheduling

- **The principle of optimum task size**: Scheduling tends to achieve maximum efficiency when the task sizes are small, and all tasks of same order of magnitude.
- **Principle of optimum production plan:** The planning should be such that it imposes an equal load on all plants.
- **Principle of optimum sequence:** Scheduling tends to achieve the maximum efficiency when the work is planned so that work hours are normally used in the same sequence.

Issues in Scheduling in Parallel Computing

- The factors those needed to be considered while scheduling in parallel computing are as follows:
- Check which task should be executed by which processor.
- Minimize total turnaround time and total response time for all the processes.
- Balance the workload on all the available processors.
- Order of execution of various queues.
- Finally join the results of all the processors and to give the consistent overall result to the user.
- Maximum and optimal use of all the resources.

TYPES OF SCHDULING

- Forward scheduling
- Forward scheduling is commonly used in job shops where customers place their orders on "needed as soon as possible" basis.
- Forward scheduling determines start and finish times of next priority job by assigning it the earliest available time slot and from that time, determines when the job will be finished in that work centre.

TYPES OF SCHDULING

- Backward scheduling
- It is often used in assembly type industries and commit in advance to specific delivery dates. Backward scheduling determines the start and finish times for waiting jobs by assigning them to the latest available time slot that will enable each job to be completed just when it is due, but done before.

Lecture 28
Synchronization

Synchronization

Definition

- Synchronization is the coordination of events to operate a system in unison (same time or rate)
- Systems that operate with all parts in synchrony are said to be synchronous or in sync-and those that are not are asynchronous.
- Achieved via clocks -processes/Resource, Concurrency, Multiple nodes

Lecture 29
CLOCK SYNCHRONIZATION

CLOCK SYNCHRONIZATION

- Every computer needs a timer mechanism (called a computer clock) to keep track of current time and also For various accounting purposes such as calculating the time spent by a process in CPU utilization, disk I/O and so on, so that the corresponding user can be charged properly.
- Accounting-resource - CPU, memory, Hard Disk
- Clocks - crystal oscillate at the same freq

Cont...

- It is the job of a distributed operating system designer to devise and use suitable algorithms for properly synchronizing the clocks of a distributed system

Lecture 30 MUTUAL EXCLUSION

Mutual Exclusion

- Several resources in a system that must not be used simultaneously by multiple processes if program operation is to be correct.
  - File must not be simultaneously updated by multiple processes-DBMS
  - Printers must be restricted to a single process at a time. -spooling-Queue
  - Processes(A, B) - Printer
- Therefore, exclusive access to such a shared resource by a process must be ensured.
- Mutual exclusion - are introduced to prevent processes from executing concurrently within their associated critical sections/resources.

Cont...

- An algorithm for implementing mutual exclusion must satisfy the following requirements:
- Issues in Recovery from Deadlock:
  - Selection of Victim(s):

  -min recovery cost, prevention of starvation
  - Use of Transaction Mechanism

- Three conditions of policy must be present for a deadlock to be possible:
  1. Mutual exclusion. Only one process may use a resource at a time. No process may access a resource unit that has been allocated to another process.
  2. Hold and wait. A process may hold allocated resources while awaiting assignment of other resources.
  3. No preemption. No resource can be forcibly removed from a process holding it.
  4. Circular wait. A closed chain of processes exists, such that each process holds at least one resource needed by the next process in the chain

Diagram

- Three conditions of policy must be present for a deadlock to be possible:
- 1. Mutual exclusion. Only one process may use a resource at a time. No process may access a resource unit that has been allocated to another process
- 2. Hold and wait. A process may hold allocated resources while awaiting assignment of other resources.
- 3. No preemption. No resource can be forcibly removed from a process holding it
- 4. Circular wait. A closed chain of processes exists, such that each process holds at least one resource needed by the next process in the chain

Selection of Victims

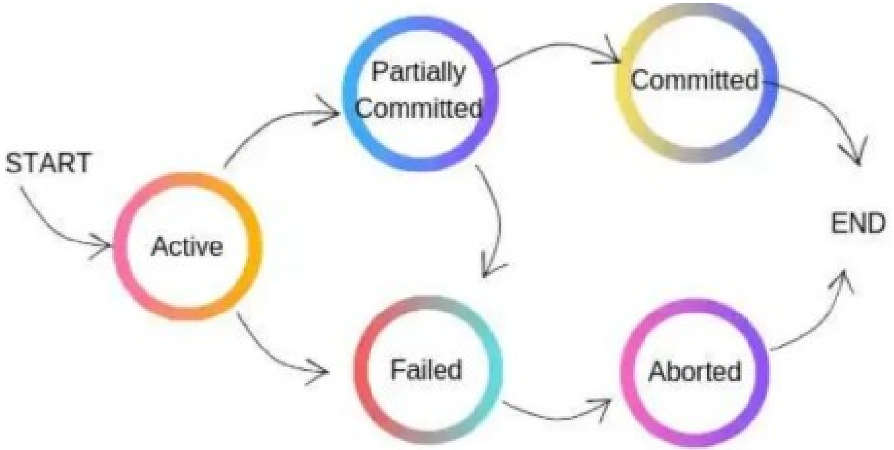
- Based on two major factors
- Minimization of recovery cost - those processes should be selected as victims whose termination / rollback will incur the minimum recovery cost.
- Some of the factors that may be considered for this purpose are
  - a) The priority of the processes; (BANK- P1-opening an account, P2-checking balance, P3-depositing)
  - b) The nature of the processes, such as interactive or batch and possibility of return with no ill effects;
  - c) The number and types of resources held by the processes;
  - d) The length of service already received and the expected length of service further needed by the processes; and e) The total number of processes that will be affected.

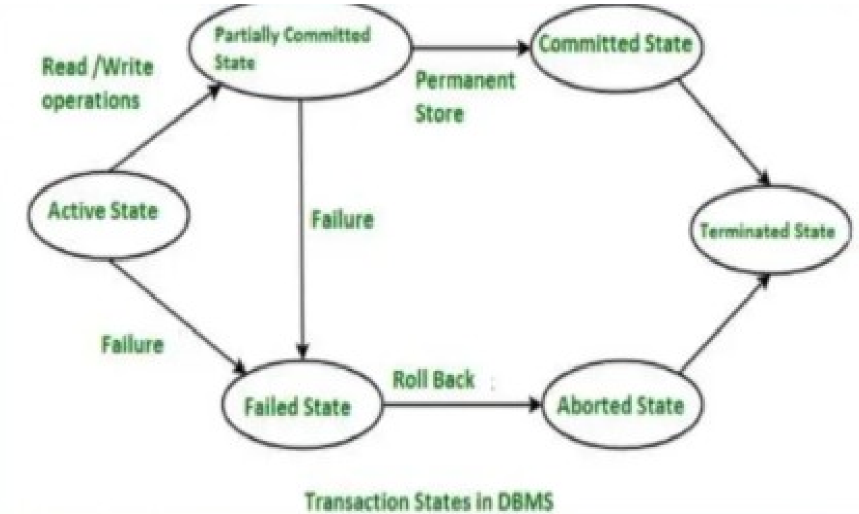
Selection of Victims

- Prevention of starvation:
- If a system only aims at minimization of recovery cost, it may happen that the same process (probably because its priority is very low) is repeatedly selected as a victim and may never complete.

Use of transaction mechanism

- The use of transaction mechanism ensure all or no effect DBMS


START
Active
Partially Committed
Committed
Failed
Aborted
END

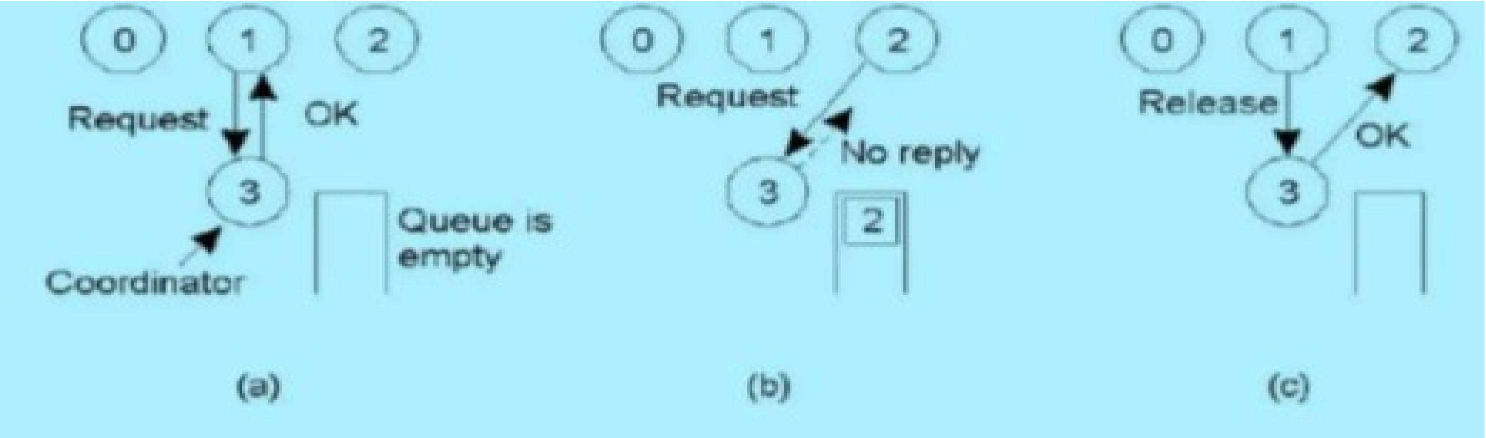

Read /Write operations
Partially Committed State
Committed State
Permanent Store
Active State
Failure
Terminated State
Failure
Failed State
Roll Back
Aborted State
Transaction States in DBMS

Centralized Algorithm

a) Process 1 asks the coordinator for permission to enter a critical region. Permission is granted.

b) Process 2 then asks permission to enter the same critical region. The coordinator does not reply.

c) When process 1 exits the critical region, it tells the coordinator, when then replies to process 2.



Advice

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.